

BrightWebApp: A Framework for Teaching Life-Cycle Assessment and Supply Chain Analysis in the Browser with WebAssembly and Brightway

Michael P. Weinold^{1,2} and Christopher Mutel³

¹ Laboratory for Energy Systems Analysis, PSI Centers for Nuclear Engineering & Sciences and Energy & Environmental Sciences, Villigen, Switzerland ² Chair of Energy Systems Analysis, Institute of Energy and Process Engineering, Department of Mechanical and Process Engineering, ETH Zurich, Zurich, Switzerland ³ Département de Sentier (DdS) Non-Profit Association, Riniken, Switzerland

DOI: 10.xxxxxx/draft

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#))

Summary

BrightWebApp is a [Python](#) software package, stand-alone web application and development template that demonstrates a framework for building complex life-cycle assessment (LCA) and supply chain analysis applications that run in the browser using [WebAssembly \(WASM\)](#). It combines the Brightway ([Mutel, 2017](#)) LCA framework for calculation with the Holoviz Panel ([Rudiger et al., 2024](#)) library for interactivity, executing entirely client-side via Pyodide ([Droettboom et al., 2024](#)). As a stand-alone web application, it allows users to perform life cycle assessment, life cycle impact assessment and supply chain analysis of the entire US economy, based on the US Environmentally-Extended Input-Output (USEEIO) database ([Yang et al., 2017](#)) directly in their web browser without the need for a local installation of either Python or Brightway. As a web application, it is therefore uniquely suited for teaching quantitative sustainability assessment without technical overhead. As a development template, it provides a starting point for developers to build their own web applications that leverage the Brightway framework in the browser and comes with extensive documentation describing the build and deployment process.

Statement of need

The Brightway framework ([Mutel, 2017](#)) is a powerful tool for life-cycle assessment and supply chain analysis, now widely used in academic and industry settings. However, it is designed primarily for use from the command-line or through local Python environments, which poses a challenge in the educational context. In life cycle assessment classes and workshops, participants often bring enterprise-managed devices where installing Python packages is restricted, forcing organizers to provision and maintain a Jupyter server and user management services. On the other hand, a standalone web application as the obvious alternative must typically use a complex backend to expose the calculation framework or rely on serving static, pre-computed data to users. A recent example of this latter static approach is the calculator application ([Sacchi et al., 2022](#)), which assesses the environmental impact of automobiles. In this context, BrightWebApp addresses the aforementioned issues by providing the first template of a WebAssembly-enabled web application for complex life cycle assessment calculations in the browser, eliminating the need for either a local Python environment or a server backend.

State of the field

Several established open-source tools address adjacent problems in life-cycle assessment and supply chain analysis. The Brightway framework itself (Mutel, 2017) provides the calculation core but is designed for local command-line use. Activity Browser (Steubing et al., 2020) offers a desktop graphical interface to Brightway but still requires a local installation of Python and the framework on each user device. openLCA (Ciroth, 2007) is a widely used Java-based LCA application with similar local-installation requirements. Domain-specific tools built on Brightway, such as carculator (Sacchi et al., 2022) for automotive LCA, have begun to expose results through web interfaces, but these typically serve only pre-computed scenarios because the underlying calculation engine cannot run in the browser.

Table 1 Selection of representative open-source software packages for life-cycle assessment and supply chain analysis. Abbreviations: Py. = Python, Jv. = Java.

Name	Lang.	Purpose
Brightway (2017)	Py.	LCA calculation framework (local)
Activity Browser (2020)	Py.	Desktop GUI for Brightway
carculator (2022)	Py.	Automotive LCA with static web frontend
openLCA (2023)	Jv.	General-purpose LCA software (local)

BrightWebApp therefore fills a novel scholarly and pedagogical niche. To the best knowledge of the authors, it is the first complete template for running a full Brightway-based LCA workflow client-side in the browser. It therefore enables sustainability education in without the need for platform-specific local Python installations or server backends.

Software design

The central design choice of BrightWebApp is to execute the full Brightway stack (Mutel, 2017) inside the user's browser through the Pyodide (Droettboom et al., 2024) WebAssembly Python runtime, rather than on a remote server. The interactive user interface is built with Holoviz Panel (Rudiger et al., 2024), whose components are rendered directly in the browser through Pyodide. This client-side architecture eliminates the need for a server backend, user authentication, or session management, and ensures that no user data leaves the device.

The package is organized as a small set of focused modules. The brightway module wraps the underlying Brightway calculation primitives (bw2data, bw2io, bw2calc) for use in the browser context. The traversal module implements supply-chain graph traversal on top of bw_graph_tools for the example use case. The modifications module exposes interactive parameter adjustments that propagate through the supply chain in real time. The same package can be used both as a stand-alone web application and as a development template that downstream projects can fork to build their own browser-based LCA tools.

Research impact statement

The framework implemented in BrightWebApp has supported scholarly and teaching outputs. It was developed in the context of the teaching requirements of the doctoral research *Hybrid Life-Cycle Assessment for Sustainable Aviation* (Weinold, 2026) and has been used to deliver hands-on life-cycle assessment teaching in the context of multiple academic life cycle assessment seminars:

- DdS/PSI Spring School 2024, Quebec (May 2024)
- DdS/PSI Autumn School (October 2024)
- DdS/PSI Summer School 2025, Singapore (June 2025)

Emission scope splitting and supply chain analysis

BrightWebApp implements an example use-case that demonstrates how to split life-cycle assessment results into different “emission scopes” according to the Greenhouse Gas Protocol accounting standard (Bhatia et al., 2004), shown in Figure 1.

Scope 1: Direct GHG emissions. These are from sources that are owned or controlled by the company, for example, emissions from combustion in owned or controlled boilers, furnaces, vehicles, etc.

Scope 2: Electricity indirect GHG emissions. These are from the generation of purchased electricity consumed by the company. Scope 2 emissions physically occur at the facility where electricity is generated.

Scope 3: Other indirect GHG emissions. All other indirect emissions.

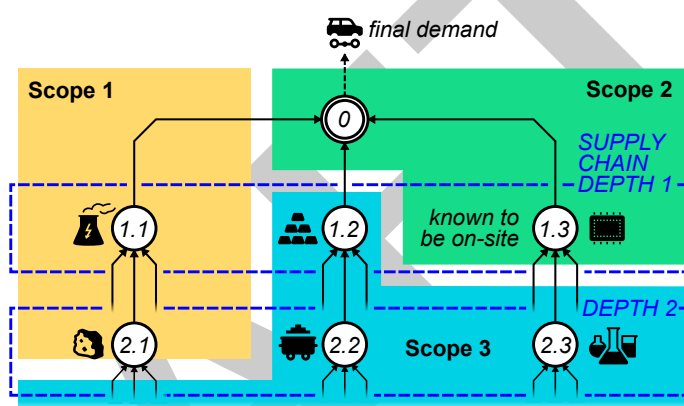


Figure 1: Diagram of the example use-case implemented in the BrightWebApp package. Here, every numbered circle represents a production node in the supply chain graph of automotive manufacturing. The supply chain can be split into different levels of “depth”. Every node can be assigned to one of three “emission scopes”, as defined by the Greenhouse Gas Protocol accounting standard (Bhatia et al., 2004)

Users can also interactively explore how changes in the efficiency of individual production processes impact the overall supply chain and associated environmental impacts. This is illustrated in Figure 2, which shows a single branch of a supply chain tree. Here, users can modify the efficiency of individual production processes (i.e. how much input is required to produce a unit of output) and immediately see how this affects the total value chain.

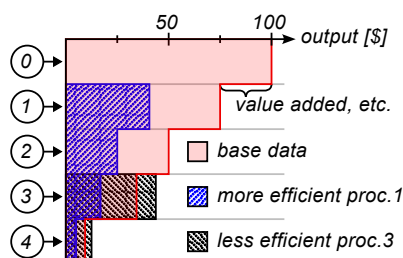


Figure 2: Diagram of a single branch in a supply chain tree in the context of the economic analysis implemented in the BrightWebApp. The diagram here shows a final demand of 100\$ worth of the product produced by sector 0. This induces demand for production in sectors upstream 1, which induces demand for production in sector 2, etc. Naturally, upstream processes contribute successively less. Red bars indicate the “base data” recorded in the economic data. Users can now change the “efficiency” (=how much upstream product is required) for individual processes. This provides instructive insights into how local efficiency changes impact the total value chain - and associated environmental or socio-economic impacts.

97 AI Usage disclosure

98 Large language models (LLMs) were used as programming assistants for repository
99 maintenance, the improvement of unit and integration tests, and the formatting of this
100 publication. All resulting changes were reviewed by the authors.

101 Acknowledgements

102 This work has been supported by the Swiss Innovation Agency Innosuisse in the context
103 of the WISER flagship project (PFFS-21-72). In addition, Michael P. Weinold gratefully
104 acknowledges the support of the Swiss Study Foundation.

105 References

- 106 Bhatia, P., Ranganathan, J., & others. (2004). *The Greenhouse Gas Protocol*. World
107 Resources Institute. ISBN: 1569735689
- 108 Ciroth, A. (2007). ICT for environment in life cycle applications openLCA—a new open
109 source software for life cycle assessment. *The International Journal of Life Cycle*
110 *Assessment*, 12(4), 209–210. <https://doi.org/10.1065/lca2007.06.337>
- 111 Droettboom, M., Yurchak, R., Chatham, H., & others. (2024). *pyodide/pyodide* (Version
112 0.29.0). Zenodo. <https://doi.org/10.5281/zenodo.5135071>
- 113 Mutel, C. (2017). Brightway: An open source framework for life cycle assessment. *Journal*
114 *of Open Source Software*, 2(12), 236. <https://doi.org/10.21105/joss.00236>
- 115 Rudiger, P., Madsen, M. S., Hansen, S. H., & others. (2024). *holoviz/panel* (Version 1.8.5).
116 Zenodo. <https://doi.org/10.5281/zenodo.3706648>
- 117 Sacchi, R., Bauer, C., Cox, B., & Mutel, C. (2022). When, where and how can the
118 electrification of passenger cars reduce greenhouse gas emissions? *Renewable and*
119 *Sustainable Energy Reviews*, 162, 112475. <https://doi.org/10.1016/j.rser.2022.112475>
- 120 Steubing, B., De Koning, D., Haas, A., & Mutel, C. L. (2020). The activity browser—an
121 open source LCA software building on top of the brightway framework. *Software*
122 *Impacts*, 3, 100012. <https://doi.org/10.1016/j.simpa.2019.100012>
- 123 Weinold, M. (2026). *Hybrid life-cycle assessment for sustainable aviation*. ETH Zurich.
124 <https://doi.org/10.3929/ethz-c-000790524>
- 125 Yang, Y., Ingwersen, W. W., Hawkins, T. R., Srocka, M., & Meyer, D. E. (2017). USEEIO:
126 A new and transparent united states environmentally-extended input-output model.
127 *Journal of Cleaner Production*, 158, 308–318. [https://doi.org/10.1016/j.jclepro.2017.](https://doi.org/10.1016/j.jclepro.2017.04.150)
128 [04.150](https://doi.org/10.1016/j.jclepro.2017.04.150)